Zihang Xiao

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Education

Harvard University — September 2025 - Present

Ph.D. Student in Program in Neuroscience (PiN)

University of California, Santa Barbara

Bachelor of Science in Neural Computation — September 2021-January 2025

- Selected Coursework: Regression Analysis, Statistical Machine Learning, Signal Analysis and Processing, Computational Neuroscience, Systems Neuroscience, Deep Learning, Non-Linear Dynamics, Circuits and Electronics

Experience

Research Assistant, University of California, Santa Barbara, Sung Soo Kim Lab — September 2021-June 2025

- Built a connectome-based ring-attractor ODE model of the Drosophila anterior visual pathway to test hypotheses about ring-neuron visual maps.
- Designed and built a low-cost, wind-based closed-loop virtual reality system (wind-powered treadmill, custom airflow control, Unity environment, flexible display) for multimodal navigation in flies.
- Trained CNN models to predict ring-neuron responses from visual inputs, used gradient-based feature optimization to infer effective receptive fields, and validated these features with two-photon imaging and behavior.

June 2023-September 2024

Summer SURF Fellow, California Institute of Technology, Elizabeth J. Hong Lab

- Built a behavioral apparatus for precise odor delivery and real-time fly position tracking, enabling quantification of odor-encountering behavior
- Wrote scripts for data cleaning, analyzing, and characterizing free fly behavior dynamics during different odors
- Conducted experiment in vivo two-photon functional calcium imaging experiments in head-fixed behaving flies to investigate early-age natural odor exposure-induced modifications in mushroom body activity.

Research Assistant, Zhejiang University, Xiao Dong Wang Lab — May 2021-September 2021

- Built a Y-maze behavioral setup with integrated two-photon imaging to study lateral entorhinal cortex (LEC) activity in mice
- Analyzed microscopy data from LEC of the mice and its behavioral data in the Y maze using PCA and SVM and discovered a subset of neurons in LEC related to anxiety behavior

Activities

Founder, Quantitative Biology Club @ UCSB — June 2023-June 2025

- Collaborate with other team members for holding and designing the guest speaker series and journal clubs about recent biological advancement and its quantitative methodology
- Recruiting faculty advisor and raise funding for the club.

Publications

Manuscripts Under Review:

- Dylla, K.V., **Xiao, Z.**, Dong, L.Y., O'Connell, T.F., Hong, E.J. "Early life experience with natural odors modifies olfactory behavior through an associative process." (Under Review at ***Current Biology***)

Manuscripts in Preparation:

- Lai, J., Garner, D., **Xiao, Z.**, Hwang, L., Roshkovan, L., Wolff, T., Rubin, G.M., Natesan, D.D., Kim, S.S. "Connectome and functional imaging reveal visual features critical for navigation in *Drosophila melanogaster*." (Manuscript in preparation)

Skills

Python, C++, C#, R, MATLAB, PyTorch, CAD Design, JavaScript, Data Analysis